

1. Let  $A = \{\text{Fish, Bird, Bee}\}$  and  $S = \{\text{California, Georgia}\}$ .

(a) One example of a function  $f : A \rightarrow S$  is

$$f(\text{Fish}) = \text{California}, \quad f(\text{Bird}) = \text{Georgia}, \quad f(\text{Bee}) = \text{California}.$$

This is a function because every element of  $A$  is assigned exactly one element of  $S$ .

(b) A relation  $R$  from  $A$  to  $S$  is any subset of  $A \times S$ . For example,

$$R = \{(\text{Fish, California}), (\text{Fish, Georgia}), (\text{Bird, California}), (\text{Bee, Georgia})\}.$$

Then  $R \subseteq A \times S$  and  $|R| = 4$ .

(c) Any function  $f : A \rightarrow S$  must have exactly one ordered pair for each element of  $A$ , so its graph has exactly  $|A| = 3$  ordered pairs. Therefore, no relation from  $A$  to  $S$  with cardinality 4 can be a function.

2. Let  $\mathbb{Z}$  denote the set of all integers, and define  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  by  $f(n) = 4n$ .

(a)  $f$  is one-to-one (injective): if  $f(a) = f(b)$ , then  $4a = 4b$ , and dividing by 4 gives  $a = b$ .

(b)  $f$  is not onto (surjective): the range of  $f$  is

$$\text{range}(f) = \{4n \mid n \in \mathbb{Z}\},$$

In Layman's terms, the set of multiples of 4. For example, we find  $1 \notin \text{range}(f)$  since there is no integer  $n$  with  $4n = 1$ .

3. Let  $A = \{a, b, c, d, e\}$  and  $B = \{a, e\}$ . We want the number of sets  $C$  such that  $C \subseteq A$  and  $B \subseteq C$ .

Since  $B \subseteq C$ , the set  $C$  must contain  $a$  and  $e$ . Since  $C \subseteq A$ , the only additional elements  $C$  can possibly include are from  $\{b, c, d\}$ . Thus  $C$  has the form

$$C = \{a, e\} \cup T \quad \text{where } T \subseteq \{b, c, d\}.$$

There are  $2^3 = 8$  choices for  $T$ , so there are 8 such sets  $C$ .

4. We want to show: if we select any 14 unique integers from  $S = \{1, 2, 3, \dots, 25\}$ , then two selected integers sum to 26.

(a) We wish to consider the 13 pigeonholes given by grouping the numbers as

$$\{1, 25\}, \{2, 24\}, \{3, 23\}, \dots, \{12, 14\}, \{13\}.$$

The pigeons are the 14 selected integers (each selected integer is placed into the unique group containing it).

- (b) We begin by placing each of the 14 selected integers into its corresponding pigeonhole above. Since there are 14 pigeons and only 13 pigeonholes, the pigeonhole principle guarantees that some pigeonhole will contain at least two selected integers.

The hole  $\{13\}$  cannot contain two selected integers because all selected integers are unique, so the hole with at least two selected integers must be one of the 12 two-element holes  $\{k, 26 - k\}$  for  $k = 1, \dots, 12$ . Therefore, we find for some  $k \in \{1, \dots, 12\}$ , both  $k$  and  $26 - k$  were selected, and hence

$$k + (26 - k) = 26.$$

So, among any 14 unique integers chosen from  $\{1, 2, \dots, 25\}$ , there exist two whose sum is 26.